Cross-Domain Feature Augmentation for Domain Generalization

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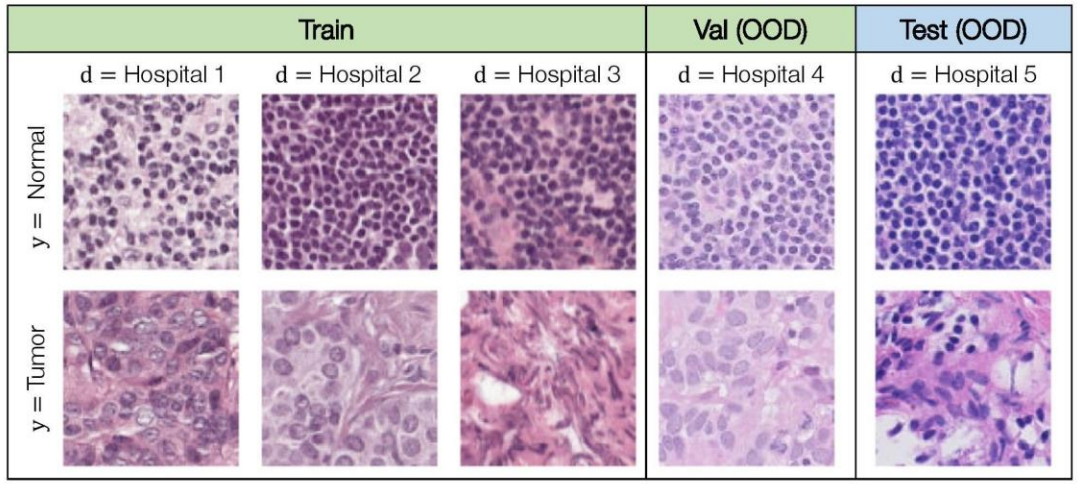
Introduction

- Machine learning model often fails when test distribution diverges from training distribution
  - e.g. tissue slides from different hospitals
- Domain generalization: learn **invariance** across domains



WILDS-Camelyon17 dataset
https://wilds.stanford.edu/

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Motivation

- Data augmentation as a commonly-used approach to learn invariance
  - e.g. colour augmentation to learn a prediction invariant to colour change
- How to learn domain invariance via data augmentation?
  - Augment **domain-specific** information
  - Rely less on information that is not general across domains

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Motivation

- Augmentation in the feature space is more versatile than input space
  - Feature semantics can be leveraged

- How to consider feature semantics?
  - Decompose feature to 4 parts

Information contains	Class-specific (Z_c)	Class-generic ($Z_{\neg c}$)
Domain-specific (Z_d)	$Z_{c,d}$	$Z_{\neg c,d}$
Domain-generic ($Z_{\neg d}$)	$Z_{c,\neg d}$	$Z_{\neg c,\neg d}$

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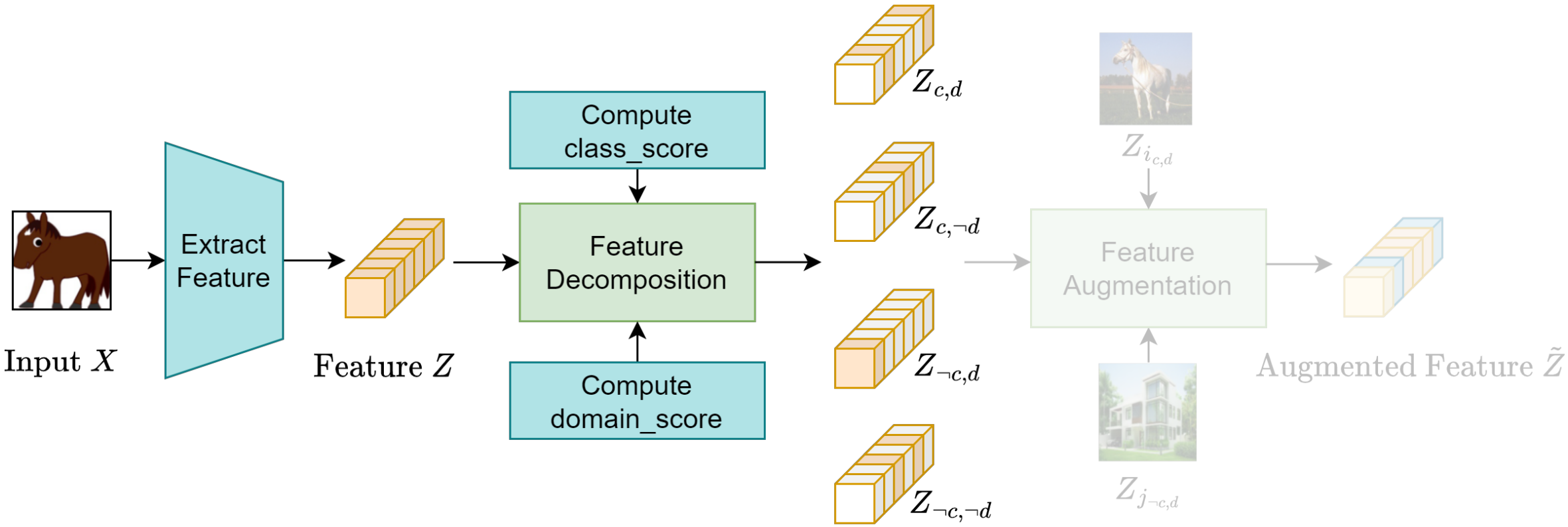
Proposed method: XDomainMix



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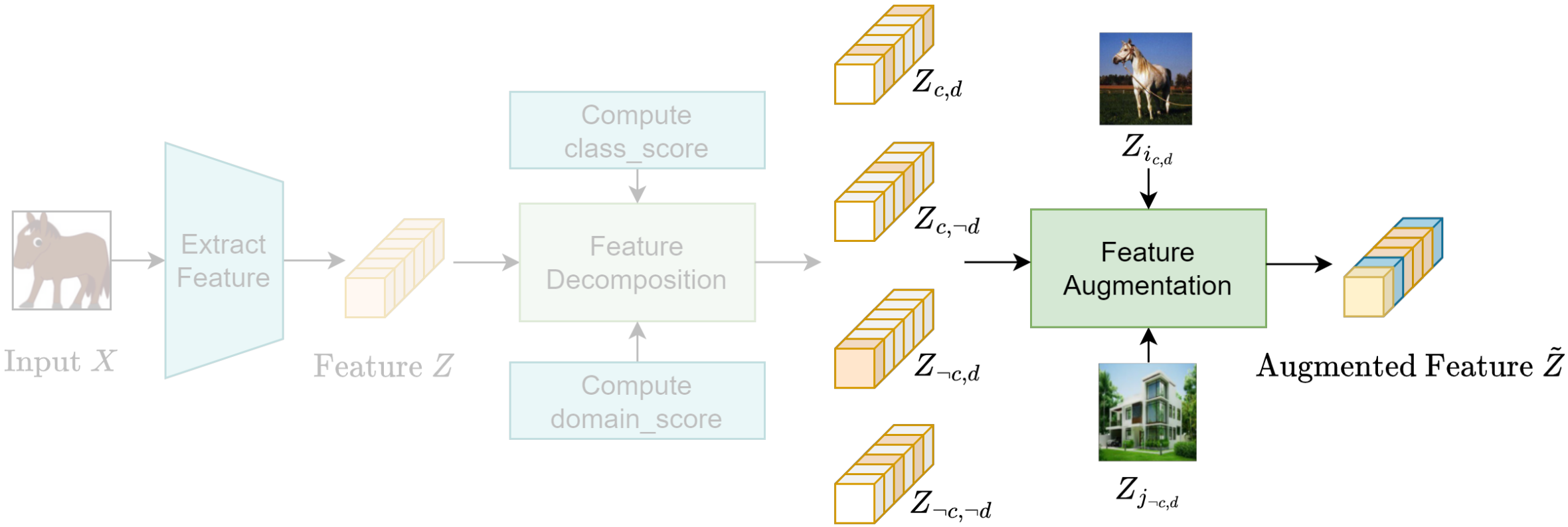

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Performance Study

- Domain generalization performance

Method	Camelyon17	FMoW	PACS	TerraIncognita	DomainNet
ERM	70.3±6.4	32.3±1.3	85.5±0.2	46.1±1.8	43.8±0.1
GroupDRO	68.4±7.3	30.8±0.8	84.4±0.8	43.2±1.1	33.3±0.2
RSC	77.0±4.9ˆ	32.6±0.5ˆ	85.2±0.9	46.6±1.0	38.9±0.5
MixStyle	62.6±6.3ˆ	32.9±0.5ˆ	85.2±0.3	44.0±0.7	34.0±0.1
DSU	69.6±6.3ˆ	32.5±0.6ˆ	85.5±0.6ˆ	41.5±0.9ˆ	42.6±0.2ˆ
LISA	77.1±6.5	35.5±0.7	83.1±0.2ˆ	47.2±1.1ˆ	42.3±0.3ˆ
Fish	74.7±7.1	34.6±0.2	85.5±0.3	45.1±1.3	42.7±0.2
XDomainMix	**80.9±3.2**	**35.9±0.8**	**86.4±0.4**	**48.2±1.3**	**44.0±0.2**

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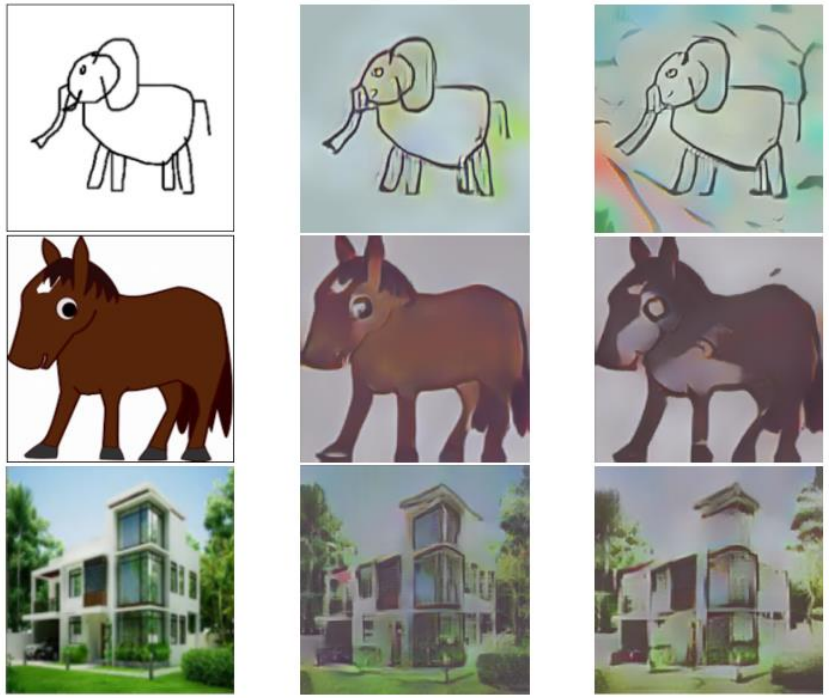
Performance Study



(a) Original (b) DSU (c) XDomainMix

Augmented feature reconstruction. XDomainMix enhances diversity while preserving class semantics.

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